Loopline P2P Application - Complete Technical Documentation

Version: 1.0
Last Updated: May 25, 2026
Author: Development Team

Table of Contents

System Overview

Loopline is a peer-to-peer (P2P) file transfer and synchronization application with a distributed architecture. It consists of:

- **Backend**: C++ server handling file transfers, HTTP API, and peer coordination
- **Frontend**: User interface for file management and peer interaction

The system enables secure file sharing between multiple peers with real-time synchronization of shared folders.

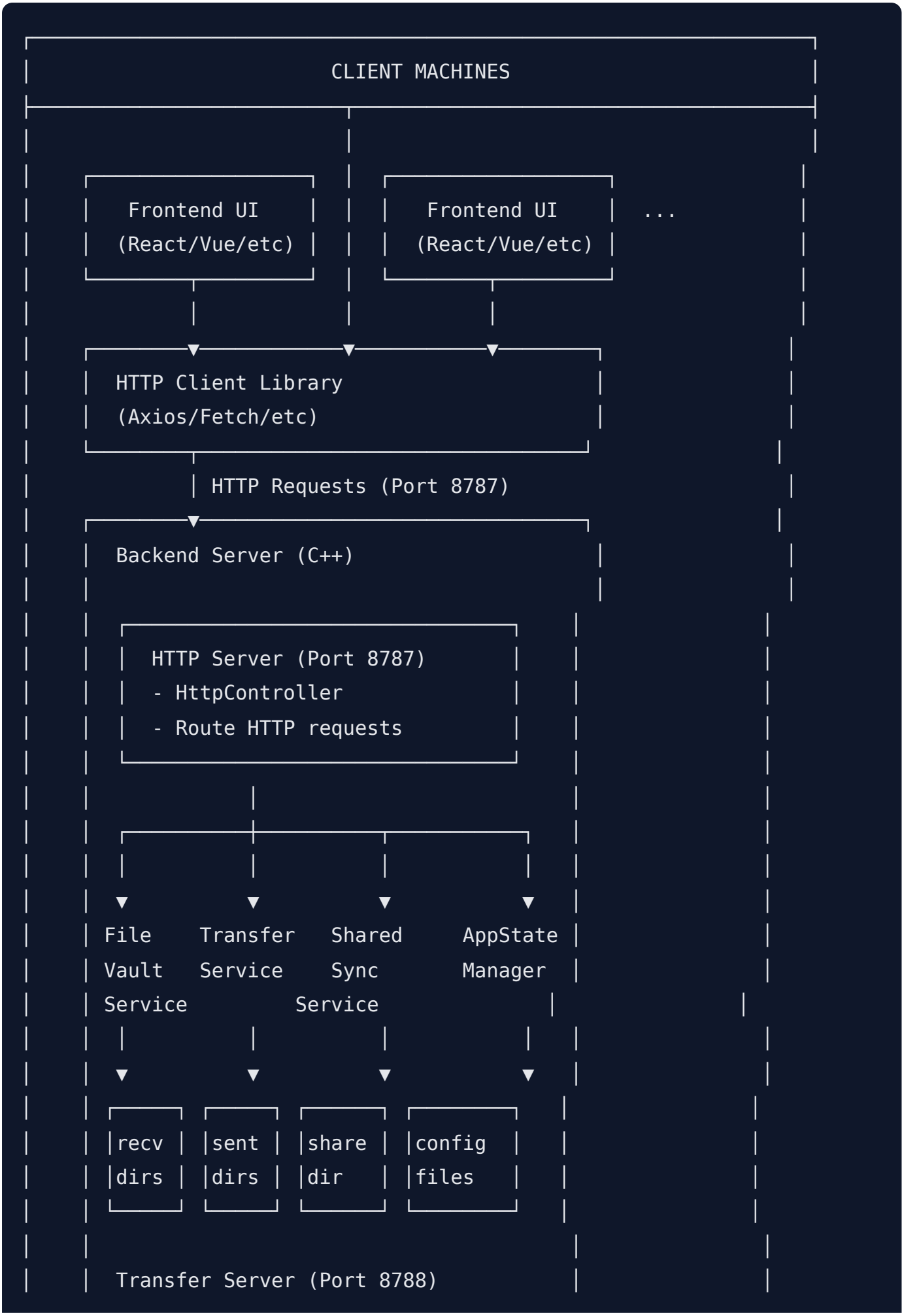
Key Features

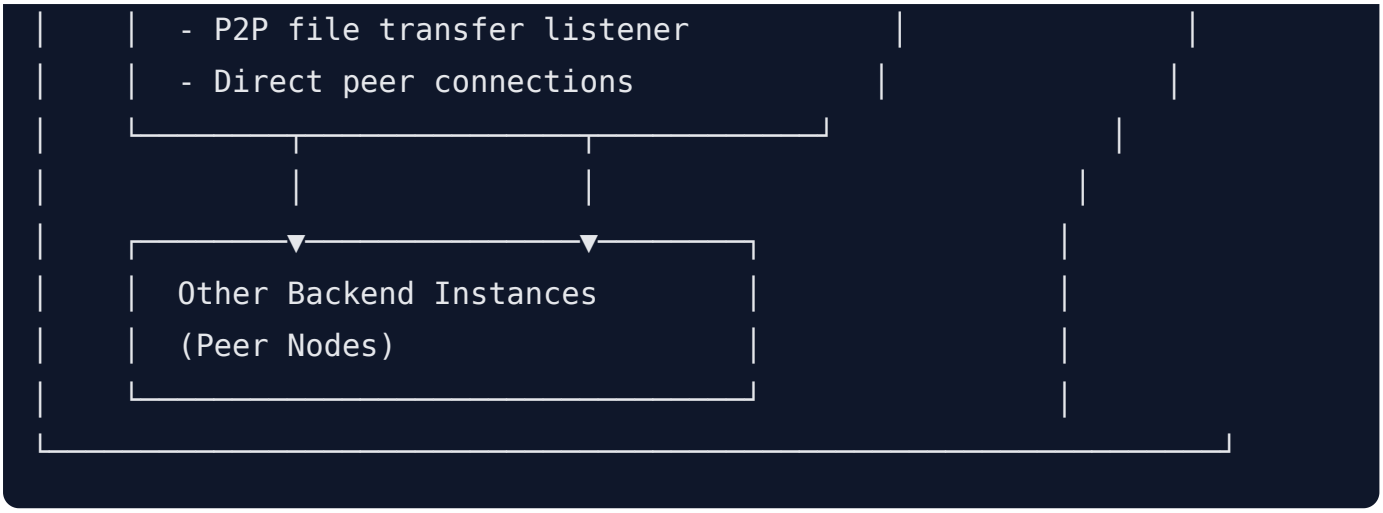
- HTTP REST API for file operations
- Direct P2P file transfer between nodes

- Real-time shared folder synchronization
- Multi-peer network support
- Cross-platform compatibility (Windows, Linux, macOS)
- Configurable network parameters
- Atomic operations for state management

Architecture

High-Level System Diagram


CLIENT MACHINES
Frontend UI
(React/Vue/etc)
Frontend UI
(React/Vue/etc)
...
HTTP Client Library
(Axios/Fetch/etc)
HTTP Requests (Port 8787)
Backend Server (C++)
HTTP Server (Port 8787)
- HttpController
- Route HTTP requests
File
Vault
Service
Transfer
Service
Shared
Sync
Service
AppState
Manager
recv
dirs
sent
dirs
share
dir
config
files
Transfer Server (Port 8788)



Architecture

Component Diagram

```
AppState (Central State)
  ├── http_port
  ├── transfer_port
  ├── bind_host
  ├── advertised_host
  ├── allow_remote_peers
  ├── directories (receive, sent, shared)
  ├── sync_peers[]
  ├── running (atomic)
  └── listener_active (atomic)

HttpServer (Main Loop)
  ├── Create Listener Socket
  ├── Setup Dependencies (Lambda Functions)
  └── Accept Loop
      ├── Accept Connection
      ├── Create Handler Thread
      └── Detach Thread

HttpController
  ├── status_json()
  ├── file_list_json()
  ├── send_file_inline()
  ├── send_file_to_peer()
  ├── sync_shared_folder_json()
  ├── save_shared_upload()
  ├── add_sync_peer()
  ├── remove_sync_peer()
  └── is_allowed_peer()

FileVaultService
  ├── file_list_json()
  ├── send_file_inline()
  └── save_shared_upload()

TransferService
  ├── transfer_listener() (Background Thread)
```

```
  ├── send_file_to_peer()
  └── receive_file()

SharedSyncService
  ├── watch_shared_folder() (Background Thread)
  └── sync_shared_folder_json()
```

Backend Documentation

Backend Class Definition

```
// filepath: backend/src/app/backend_app.hpp
#pragma once

class BackendApp {
 public:
  int run(int argc, char* argv[]);
};
```

Purpose: Entry point for the backend application

Method:

- `run(argc, argv)` - Main application entry, initializes and starts all services

File Structure

```
backend/
├── src/
│   ├── app/
│   │   ├── backend_app.hpp         (Main class definition)
│   │   └── backend_app.cpp         (Main implementation)
│   ├── controllers/
│   │   └── http_controller.hpp     (Request routing)
│   ├── models/
│   │   └── app_state.hpp           (State management)
│   ├── services/
│   │   ├── file-vault/
│   │   │   └── file_vault_service.hpp
│   │   ├── net-io/
│   │   │   └── net_io.hpp          (Socket operations)
│   │   ├── shared-sync/
│   │   │   └── shared_sync_service.hpp
│   │   └── transfer/
│   │       └── transfer_service.hpp (P2P transfers)
│   └── views/
│       └── http_view.hpp           (HTTP responses)
├── received/                        (Downloaded files)
└── sent/                           (Uploaded files)
```

Configuration

Environment Variables

Variable	Type	Default	Description
`P2P_HTTP_PORT`	int	8787	HTTP API server port
`P2P_TRANSFER_PORT`	int	8788	P2P transfer server port
`P2P_BIND_HOST`	string	0.0.0.0	Network interface to bind
`P2P_ADVERTISED_HOST`	string	{bind_host}	Host advertised to peers
`P2P_ALLOW_REMOTE`	bool	true	Allow remote peer connections

`P2P_RECEIVE_DIR`	path	./backend/received	Downloaded files directory
`P2P_SENT_DIR`	path	./backend/sent	Uploaded files directory
`P2P_SHARED_DIR`	path	./shared	Shared folder for sync
`P2P_SYNC_PEERS`	string	(empty)	Initial peers (comma/semicolon-separated)

Command-Line Arguments

Arguments override environment variables with higher priority:

```
./backend_app \
  --http 8787 \
  --transfer 8788 \
  --bind 0.0.0.0 \
  --advertise 127.0.0.1
```

Argument	Type	Example	Description
`--http`	int	8787	HTTP port override
`--transfer`	int	8788	Transfer port override
`--bind`	string	0.0.0.0	Bind interface override
`--advertise`	string	192.168.1.100	Public host override

Backend Components

1. AppState Model

Centralized application state container.

Responsibilities:

- Store configuration values
- Manage peer list
- Maintain server status flags
- Generate status JSON

Key Members:

```
int http_port;
int transfer_port;
std::string bind_host;
std::string advertised_host;
bool allow_remote_peers;
std::filesystem::path receive_dir;
std::filesystem::path sent_dir;
std::filesystem::path shared_dir;
std::atomic<bool> running;
std::atomic<bool> listener_active;
std::vector<transfer::PeerEndpoint> sync_peers;
```

Methods:

- `status_json()` - Returns JSON representation of state
- `add_sync_peer(PeerEndpoint)` - Add peer to network
- `remove_sync_peer(PeerEndpoint)` - Remove peer from network

---

2. FileVaultService

Manages file storage and retrieval operations.

Responsibilities:

- List files by category (received, sent, shared)
- Send files to HTTP clients
- Handle file uploads
- Generate file list JSON

Methods:

- `file_list_json(kind)` - Get files of type (received/sent/shared)
- `send_file_inline(socket, kind, name)` - Stream file to HTTP client
- `save_shared_upload(socket, name, body)` - Store uploaded file
- `send_file_to_peer(socket, host, port, name, body)` - Forward to peer

Directory Structure:

```
backend/
├── received/        (Downloaded from peers)
├── sent/            (Uploaded by this node)
└── shared/          (Synchronized across network)
```

---

3. TransferService

Handles peer-to-peer file transfers.

Responsibilities:

- Listen for incoming P2P connections (port 8788)
- Receive files from remote peers
- Send files to remote peers
- Manage peer connections

Methods:

- `transfer_listener()` - Accept incoming peer connections
- `send_file_to_peer(...)` - Send file to specific peer
- `receive_file(socket, name)` - Handle file from peer

Peer Connection Flow:

```
1. Client connects to transfer port
2. Service receives peer info
3. File data is transmitted
4. File stored in received directory
5. Connection closed
```

---

4. SharedSyncService

Monitors and synchronizes shared folder.

Responsibilities:

- Watch shared folder for changes
- Detect new/modified/deleted files
- Generate sync state JSON
- Notify other peers of changes

Methods:

- `watch_shared_folder()` - Main monitoring loop
- `sync_shared_folder_json()` - Get current state

Watched Events:

- File creation
- File modification
- File deletion

---

5. HttpController

Routes HTTP requests to appropriate handlers.

Dependencies (Passed via Lambda Closures):

```
struct Dependencies {
  std::function<std::string()> status_json;
  std::function<std::string(std::string)> file_list_json;
  std::function<void(socket_t, std::string, std::string)>
send_file_inline;
  std::function<void(socket_t, std::string, int, std::string,
std::vector<char>)> send_file_to_peer;
  std::function<std::string()> sync_shared_folder_json;
  std::function<void(socket_t, std::string, std::vector<char>)>
save_shared_upload;
  std::function<int()> transfer_port;
  std::function<bool()> listener_active;
  std::function<void(transfer::PeerEndpoint)> add_sync_peer;
  std::function<void(transfer::PeerEndpoint)> remove_sync_peer;
  std::function<bool(std::string)> is_allowed_peer;
};
```

Request Flow:

```
1. Accept HTTP connection
2. Parse HTTP request
3. Route to handler
4. Call appropriate dependency lambda
5. Format response with HttpView
6. Send response
7. Close connection
```

Utility Functions

`lower_copy(std::string value) → std::string`

Converts ASCII uppercase letters to lowercase.

Usage: Environment variable case-insensitive parsing

Example:

```
lower_copy("TRUE") → "true"
lower_copy("Yes") → "yes"
```

`parse_int(const std::string& value) → std::optional<int>`

Parses string to integer using `std::from_chars`.

Validation:

- Entire string must be valid integer
- No partial matches accepted

Usage: Port number parsing, configuration values

Example:

```
parse_int("8787") → 8787
parse_int("invalid") → std::nullopt
```

`env_value(const char* name, const std::string& fallback) → std::string`

Retrieves environment variable with fallback.

Logic:

1. Call `std::getenv(name)`
2. If NULL or empty, return fallback
3. Otherwise return environment value

Usage: Configuration loading

Example:

```
env_value("P2P_HTTP_PORT", "8787") → "8787" (or env value if set)
```

`env_flag(const char* name, bool fallback) → bool`

Parses environment variable as boolean flag.

Recognized True Values: "1", "true", "yes", "on" (case-insensitive)

Usage: Feature flags, boolean configuration

Example:

```
env_flag("P2P_ALLOW_REMOTE", true) → true if env="1" or "true"
```

`arg_value(int argc, char* argv[], const std::string& name, int fallback) → int`

Retrieves integer command-line argument.

Lookup Strategy:

1. Find argv[index] == name
2. Parse argv[index+1] as integer
3. Return parsed value or fallback

Usage: Port overrides, numeric configuration

Example:

```
Command: ./app --http 9999
arg_value(..., "--http", 8787) → 9999
```

`arg_string(int argc, char* argv[], const std::string& name, const std::string& fallback) → std::string`

Retrieves string command-line argument.

Lookup Strategy:

1. Find argv[index] == name
2. Return argv[index+1]
3. Return fallback if not found

Usage: Host configuration

Example:

```
Command: ./app --bind 192.168.1.1
arg_string(..., "--bind", "0.0.0.0") → "192.168.1.1"
```

`add_sync_peers_from_list(AppState& state, const std::string& raw_peers) → void`

Parses peer list and registers them.

Peer Format: `host:port` separated by comma or semicolon

Validation:

- Parse endpoint using `transfer::parse_peer_endpoint()`
- Validate with `transfer::is_allowed_peer()`
- Only add if both pass

Delimiter Priority: Semicolon > Comma

Example:

```
Input: "peer1.com:8788,peer2.com:8788;192.168.1.5:8788"
Result: Three peers registered
```

`http_server(AppState&, FileVaultService&, TransferService&, SharedSyncService&) → void`

Main HTTP server implementation.

Execution Flow:

1. Socket Creation

- Create listener socket on `bind_host:http_port`
- Log error if creation fails, return

2. Dependency Setup

- Create lambdas for each controller dependency
- Capture services by reference

3. Controller Initialization

- Create shared HttpController instance
- Pass dependencies to controller

4. Log Startup

- Print HTTP API URL to console

5. Accept Loop

- While `state.running` is true:
  - Accept incoming connection
  - Create client handler thread
  - Detach thread (fire-and-forget)
  - Continue accepting

6. Cleanup

- Close listener socket on exit

Thread Model:

- One thread per accepted connection
- Detached threads (independent lifecycle)
- No explicit thread joining

Main Application Flow

`BackendApp::run(int argc, char* argv[]) → int`

Step-by-Step Execution:

Windows Initialization (Windows Only)

```
WSADATA data{};
WSAStartup(MAKEWORD(2, 2), &data);
```

- Initialize Windows Sockets API
- Required on Windows platforms
- Cleaned up at exit with `WSACleanup()`

Configuration Loading

Priority Order:

1. Command-line arguments (highest priority)
2. Environment variables
3. Hardcoded defaults (lowest priority)

Configuration Chain:

```
state.http_port = arg_value(argc, argv, "--http",
  parse_int(env_value("P2P_HTTP_PORT", "8787")).value_or(8787));

// This evaluates as:
// 1. Check --http argument
// 2. If not found, check P2P_HTTP_PORT env
// 3. If not found, use "8787"
```

Loaded Configuration:

```
http_port        : Command-line or ENV or 8787
transfer_port    : Command-line or ENV or 8788
bind_host        : Command-line or ENV or "0.0.0.0"
advertised_host  : Command-line or ENV or bind_host
allow_remote     : ENV or true
receive_dir      : ENV or ./backend/received
sent_dir         : ENV or ./backend/sent
shared_dir       : ENV or ./shared
sync_peers       : Parsed from ENV P2P_SYNC_PEERS
```

Address Adjustment

```
if (state.advertised_host == "0.0.0.0") {
  state.advertised_host = "127.0.0.1";
}
```

- Wildcard bind address cannot be advertised
- Automatically changes to localhost

Directory Creation

```
std::filesystem::create_directories(state.receive_dir);
std::filesystem::create_directories(state.sent_dir);
std::filesystem::create_directories(state.shared_dir);
```

- Creates all required directories
- Creates parent directories if needed
- Silently succeeds if already exists

Service Initialization

```
FileVaultService file_vault_service(state);
SharedSyncService shared_sync_service(state);
TransferService transfer_service(state, shared_sync_service);
```

- Create service instances
- Pass state for configuration

- TransferService needs SharedSyncService

Background Threads

Transfer Listener Thread:

```
std::thread receiver(&TransferService::transfer_listener,
&transfer_service);
receiver.detach();
```

- Listens on transfer_port for P2P connections
- Runs independently
- Detached (not joined)

Shared Folder Watcher Thread:

```
std::thread shared_watcher(&SharedSyncService::watch_shared_folder,
&shared_sync_service);
shared_watcher.detach();
```

- Monitors shared_dir for changes
- Runs independently
- Detached (not joined)

Console Output

```
Loopline P2P receiver: 127.0.0.1:8788
Bind host: 0.0.0.0 | remote peers: enabled
Received files: /path/to/backend/received
Sent files: /path/to/backend/sent
Shared folder: /path/to/shared
HTTP API: http://127.0.0.1:8787
```

Main Loop

```
http_server(state, file_vault_service, transfer_service,
shared_sync_service);
```

- Blocking call, runs until shutdown
- Accepts HTTP connections
- Spawns handler threads

Cleanup

```
#ifdef _WIN32
  WSACleanup();
#endif
return 0;
```

- Windows Sockets cleanup
- Return success code

Frontend Documentation

Purpose

The frontend provides a user-friendly interface for:

- Viewing available files
- Uploading files to peers
- Downloading files from peers
- Managing peer connections
- Monitoring shared folder synchronization

Technology Stack Expectations

The frontend should connect to the backend HTTP API (port 8787) and implement:

- **Framework:** React, Vue, Angular, or similar SPA framework
- **HTTP Client:** Axios, Fetch API, or similar
- **UI Components:** Material-UI, Bootstrap, or custom CSS
- **State Management:** Redux, Vuex, Context API, or similar
- **Build Tool:** Webpack, Vite, or similar

Frontend Architecture

```
frontend/
├── public/
│   └── index.html
├── src/
│   ├── components/
│   │   ├── FileList.tsx          (Display files)
│   │   ├── PeerManager.tsx       (Add/remove peers)
│   │   ├── SharedFolderStatus.tsx (Sync status)
│   │   └── StatusPanel.tsx       (System status)
│   ├── services/
│   │   └── api.ts                (HTTP client)
│   ├── pages/
│   │   ├── Dashboard.tsx         (Main page)
│   │   ├── Downloads.tsx         (Received files)
│   │   ├── Uploads.tsx           (Sent files)
│   │   └── Shared.tsx            (Shared folder)
│   ├── hooks/
│   │   ├── useFileList.ts        (File fetching)
│   │   ├── usePeers.ts           (Peer management)
│   │   └── useStatus.ts          (Status polling)
│   ├── types/
│   │   └── api.ts                (TypeScript interfaces)
│   └── App.tsx                   (Root component)
└── package.json
```

Frontend Components

1. API Service Module

Handles all HTTP communication with backend.

Base URL: `http://localhost:8787` (configurable)

Methods:

```
// Get system status
getStatus(): Promise<StatusData>

// File operations
getReceivedFiles(): Promise<FileList>
getSentFiles(): Promise<FileList>
getSharedFiles(): Promise<FileList>

// File transfer
downloadFile(kind: string, filename: string): Promise<Blob>
uploadFile(filename: string, file: File): Promise<Response>
sendFileToPeer(host: string, port: number, filename: string, file:
File): Promise<Response>

// Peer management
getPeers(): Promise<PeerList>
addPeer(host: string, port: number): Promise<Response>
removePeer(host: string, port: number): Promise<Response>

// Shared folder
getSharedFolderStatus(): Promise<SyncStatus>
```

---

2. Status Panel Component

Displays real-time system information.

Displays:

- HTTP server status and address
- Transfer server status and address
- Bind host configuration
- Remote peers allowed status
- Connected peers count
- Uptime/Last sync time

Refresh Rate: 5-10 seconds polling

Data Source: `GET /api/status`

---

3. File List Component

Shows available files in categories.

Tabs/Views:

- **Received:** Files downloaded from peers
- **Sent:** Files uploaded by this node
- **Shared:** Files in synchronized folder

For Each File:

- Filename
- File size
- Modified date
- Action buttons (download/delete/share)

Data Source:

- `GET /api/files/received`
- `GET /api/files/sent`
- `GET /api/files/shared`

Upload Functionality:

- Drag-and-drop file upload
- File browser selection
- Progress indicator
- Success/error notification

Download Functionality:

- Click to download
- Shows progress
- Browser default download handler

---

4. Peer Manager Component

Manages peer connections.

Display:

- List of connected peers
- Peer host/port
- Last sync time
- Connection status

Actions:

- Add new peer (input host:port)
- Remove peer (confirmation)
- Manual sync trigger

Data Source:

- `GET /api/peers`
- `POST /api/peers` (add)
- `DELETE /api/peers/:id` (remove)

Input Validation:

- Host format validation (IP or domain)
- Port range validation (1-65535)
- Duplicate check

---

5. Shared Folder Sync Component

Displays synchronized folder status.

Displays:

- Last sync time
- Sync status (synced/syncing/error)
- File count in shared folder
- Recent changes log
- Sync health indicator

Data Source: `GET /api/shared/status`

Refresh Rate: 5 seconds polling

File List:

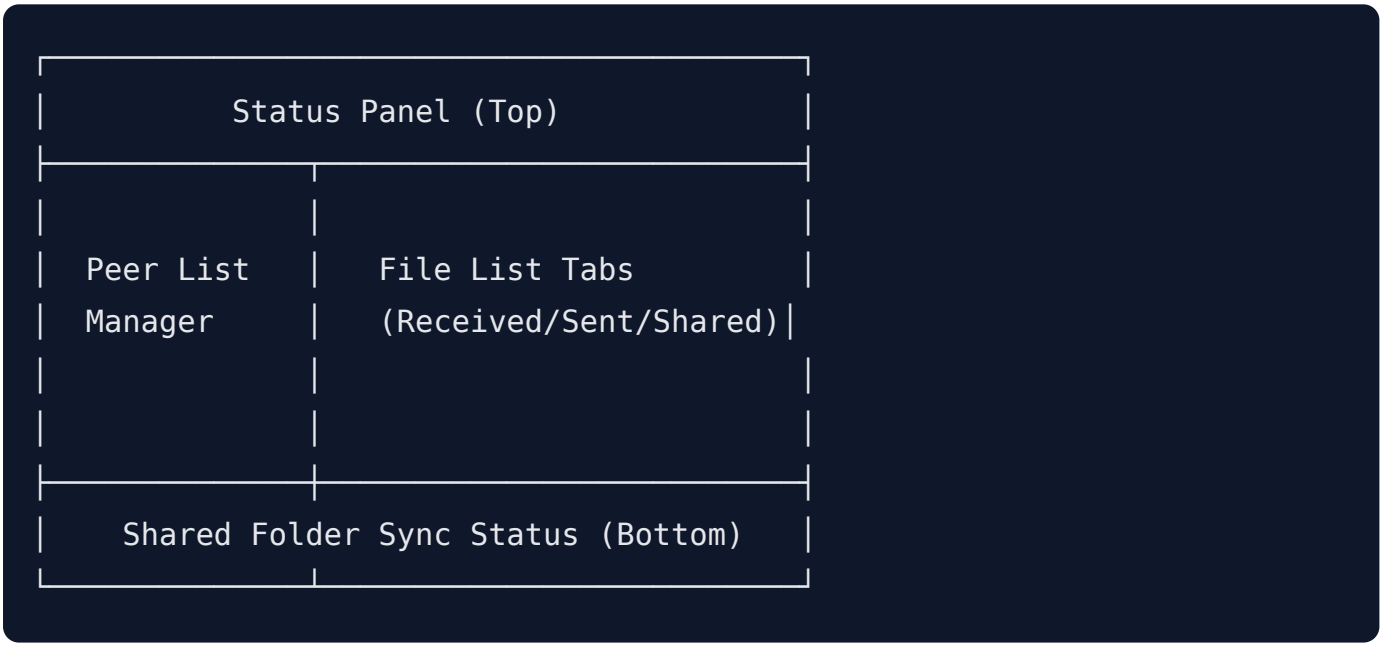
- All files in shared folder
- Size
- Last modified
- Sync state

---

6. Dashboard Page

Main landing page combining all components.

Layout:



Frontend State Management

Key State Values

```
interface AppState {
  // Status
  serverStatus: StatusData
  isConnected: boolean
  lastUpdate: Date

  // Files
  receivedFiles: File[]
  sentFiles: File[]
  sharedFiles: File[]
  loadingFiles: boolean

  // Peers
  peers: PeerEndpoint[]
  selectedPeer: PeerEndpoint | null
  loadingPeers: boolean

  // Shared Sync
  syncStatus: SyncStatus
  lastSyncTime: Date
  syncEnabled: boolean

  // UI State
  currentTab: 'received' | 'sent' | 'shared'
  uploadProgress: number
  notifications: Notification[]
}
```

State Update Triggers

- **Auto-refresh:** Status/file list/peer list every 5-10 seconds
- **User action:** File upload/download, peer add/remove
- **WebSocket (Optional):** Real-time updates for better UX
- **Manual refresh:** User-triggered force refresh

Frontend User Workflows

Workflow 1: Download File from Peer

```
1. User selects "Received Files" tab
2. Frontend fetches received file list (GET /api/files/received)
3. User clicks download button on file
4. Frontend initiates download (GET /api/files/received/{filename})
5. Browser saves file
6. User sees success notification
```

Workflow 2: Upload File

```
1. User drags file to upload zone
2. Frontend validates file
3. User confirms upload
4. Frontend sends file (POST /api/files/upload)
5. Progress bar shows transfer status
6. Success notification shown
7. File appears in "Sent Files" tab after refresh
```

Workflow 3: Add New Peer

```
1. User clicks "Add Peer" button in Peer Manager
2. Modal/form appears asking for host and port
3. User enters peer address (e.g., 192.168.1.5:8788)
4. Frontend validates input
5. User confirms
6. Frontend sends request (POST /api/peers)
7. Peer appears in peer list
8. Sync begins automatically
```

Workflow 4: Monitor Shared Folder Sync

```
1. User navigates to Shared folder tab
2. Frontend shows shared folder sync status
3. Frontend displays all files in shared folder
4. Auto-refresh shows updates every 5 seconds
5. Green indicator = synced, orange = syncing, red = error
6. User can see last sync time and file count
```

Workflow 5: Send File to Specific Peer

```
1. User selects file from Sent Files
2. User clicks "Send to Peer"
3. Peer selection dropdown appears
4. User selects destination peer
5. User confirms
6. Frontend sends request (POST /api/files/send-to-peer)
7. Progress shown
8. Success notification
```

Frontend Configuration

Environment Variables

```
REACT_APP_BACKEND_URL=http://localhost:8787
REACT_APP_POLL_INTERVAL=5000
REACT_APP_SOCKET_URL=ws://localhost:8789
REACT_APP_LOG_LEVEL=info
```

Local Storage

```
// User preferences
localStorage.setItem('theme', 'dark' | 'light')
localStorage.setItem('autoRefresh', true | false)
localStorage.setItem('pollInterval', 5000)

// Backend connection
localStorage.setItem('backendUrl', 'http://localhost:8787')
localStorage.setItem('lastConnected', timestamp)
```

API Endpoints

Status Endpoints

GET /api/status

```
Response:
{
  "http_port": 8787,
  "transfer_port": 8788,
  "bind_host": "0.0.0.0",
  "advertised_host": "127.0.0.1",
  "allow_remote_peers": true,
  "running": true,
  "listener_active": true,
  "sync_peers": [
    {"host": "peer1.com", "port": 8788},
    {"host": "192.168.1.5", "port": 8788}
  ]
}
```

GET /api/shared/sync-status

```
Response:
{
  "sync_status": "synced",
  "last_sync_time": "2026-05-25T12:05:00Z",
  "file_count": 15,
  "total_size": 104857600,
  "synced_peers": 2,
  "error_message": null
}
```

File Endpoints

GET /api/files/received

```
Response:
{
  "files": [
    {"name": "document.pdf", "size": 2048576, "modified": "2026-05-
25T10:30:00Z"},
    {"name": "image.jpg", "size": 1024000, "modified": "2026-05-
25T09:15:00Z"}
  ]
}
```

GET /api/files/sent

```
Response:
{
  "files": [
    {"name": "presentation.pptx", "size": 5242880, "modified": "2026-05-
24T14:00:00Z"}
  ]
}
```

GET /api/files/shared

```
Response:
{
  "files": [
    {"name": "shared_doc.docx", "size": 1048576, "modified": "2026-05-
25T11:45:00Z"}
  ]
}
```

POST /api/files/upload

```
Request: multipart/form-data with file
Response: {"status": "success", "filename": "uploaded_file.txt"}
```

GET /api/files/{kind}/{filename}

```
Response: File binary data
Headers: Content-Type, Content-Length, Content-Disposition
```

POST /api/files/send-to-peer

```
Request:
{
  "host": "192.168.1.5",
  "port": 8788,
  "filename": "document.pdf"
}

Response:
{
  "status": "success",
  "transfer_id": "abc123"
}
```

Peer Endpoints

GET /api/peers

```
Response:
{
  "peers": [
    {"id": 1, "host": "peer1.com", "port": 8788, "last_sync": "2026-05-25T12:00:00Z"},
    {"id": 2, "host": "192.168.1.5", "port": 8788, "last_sync": "2026-05-25T11:55:00Z"}
  ]
}
```

POST /api/peers

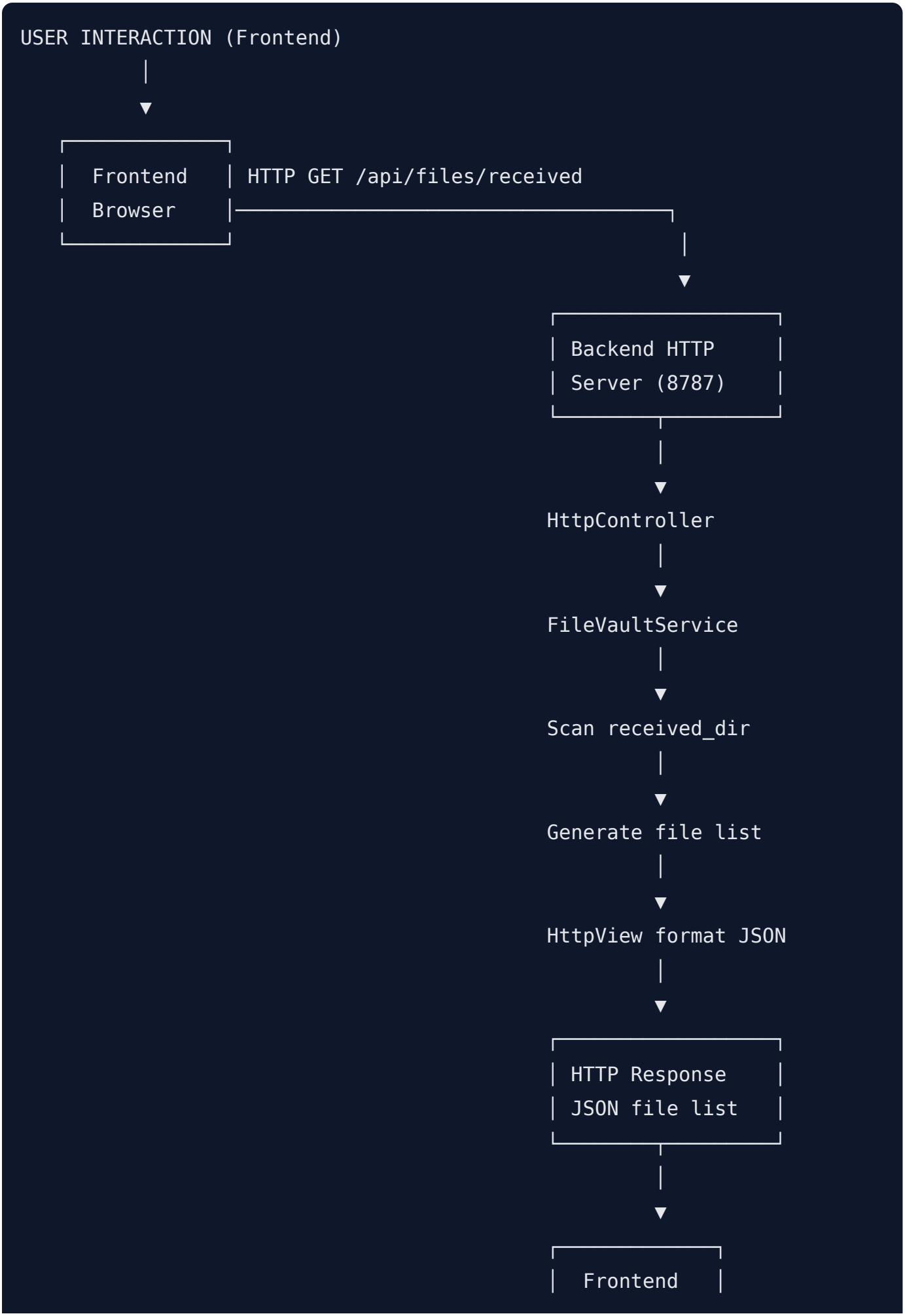
```
Request:
{
  "host": "new-peer.com",
  "port": 8788
}

Response:
{
  "status": "success",
  "peer_id": 3
}
```

DELETE /api/peers/{id}

```
Response:
{
  "status": "success",
  "removed_peer_id": 3
}
```

Data Flow

Complete Request-Response Cycle

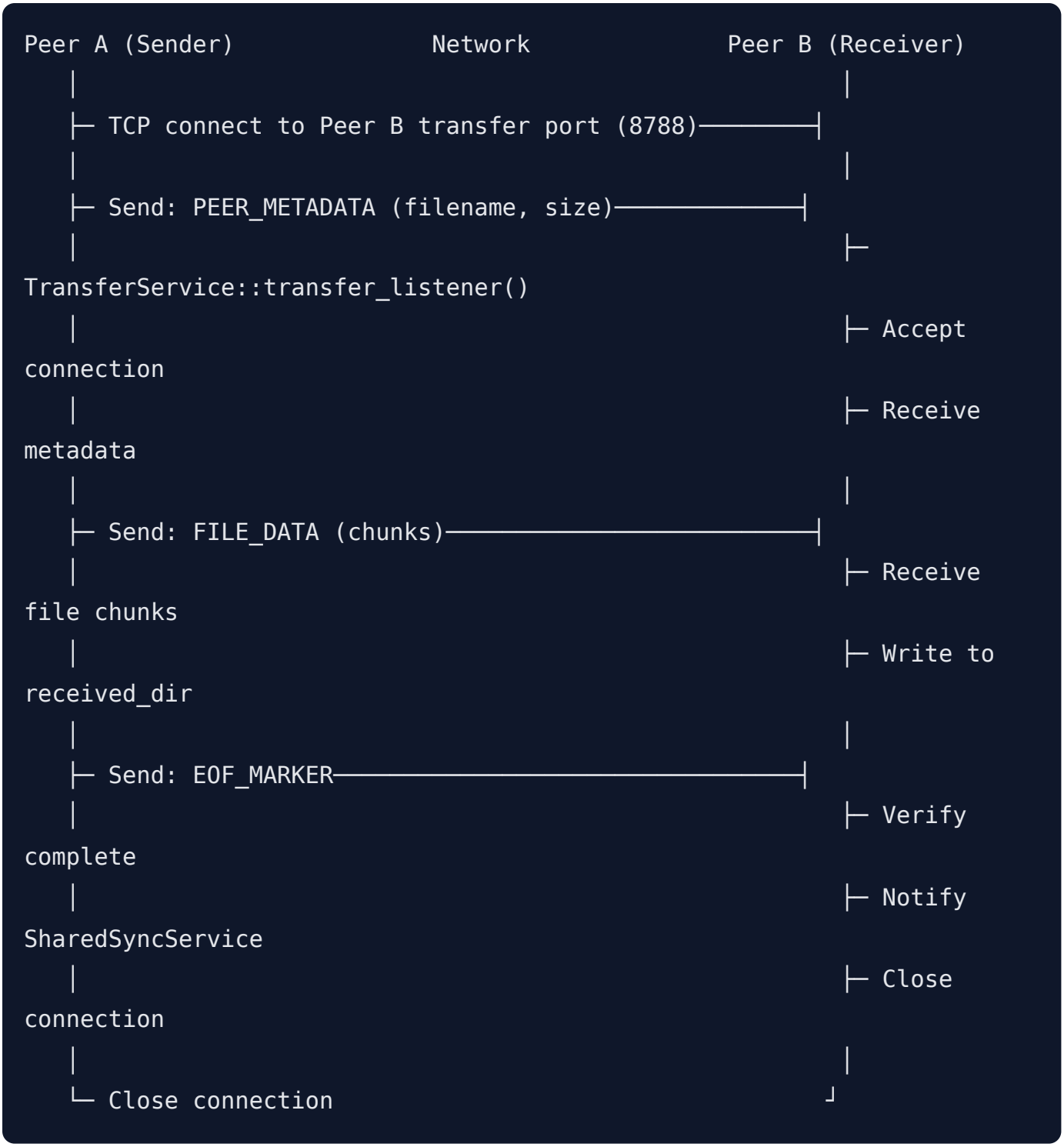

USER INTERACTION (Frontend)
Frontend Browser
HTTP GET /api/files/received
Backend HTTP Server (8787)
HttpController
FileVaultService
Scan received_dir
Generate file list
HttpView format JSON
HTTP Response JSON file list
Frontend

```
                                          │  Parse JSON │
                                          │  Render UI  │
                                          └─────────────┘
```

File Download Flow

```
Frontend (Request)
   │
   ├─ GET /api/files/received/document.pdf
   │
Backend HTTP Handler
   │
   ├─ Authenticate request
   ├─ FileVaultService::send_file_inline()
   │  │
   │  ├─ Open file from received_dir
   │  ├─ Check file exists
   │  ├─ Send HTTP headers (Content-Type, Content-Length)
   │  └─ Stream file contents
   │
   ├─ HttpView::format_response()
   │
Frontend (Response)
   │
   ├─ Receive file binary data
   ├─ Browser handles download
   └─ File saved to user's Downloads
```

P2P File Transfer Flow



Shared Folder Synchronization Flow

```
SharedSyncService
   │
   ├─ watch_shared_folder() loop (every N seconds)
   │
   ├─ Scan shared_dir for files
   │
   ├─ Detect changes:
   │  │
   │  ├─ New files: Add to peer queue
   │  ├─ Modified files: Mark for resync
   │  └─ Deleted files: Remove from peer list
   │
   ├─ For each connected peer:
   │  │
   │  ├─ Connect to transfer port
   │  ├─ Send changed files
   │  └─ Close connection
   │
   └─ Update sync_status
```

Configuration

Complete Configuration Example

Environment Setup

```
# File: .env or environment variables

# HTTP Server
P2P_HTTP_PORT=8787
P2P_BIND_HOST=0.0.0.0
P2P_ADVERTISED_HOST=192.168.1.100

# Transfer Server
P2P_TRANSFER_PORT=8788
P2P_ALLOW_REMOTE=true

# Directories
P2P_RECEIVE_DIR=/var/loopline/received
P2P_SENT_DIR=/var/loopline/sent
P2P_SHARED_DIR=/var/loopline/shared

# Initial Peers
P2P_SYNC_PEERS=peer1.company.com:8788;peer2.company.com:8788;192.168.1.5
:8788
```

Command-Line Startup

```
# Windows
.\backend_app.exe ^
  --http 8787 ^
  --transfer 8788 ^
  --bind 0.0.0.0 ^
  --advertise 192.168.1.100

# Linux/macOS
./backend_app \
  --http 8787 \
  --transfer 8788 \
  --bind 0.0.0.0 \
  --advertise 192.168.1.100
```

Docker Configuration Example

```
FROM ubuntu:22.04

WORKDIR /app

COPY backend_app .
COPY shared /shared

ENV P2P_HTTP_PORT=8787
ENV P2P_TRANSFER_PORT=8788
ENV P2P_BIND_HOST=0.0.0.0
ENV P2P_ADVERTISED_HOST=loopline.example.com

EXPOSE 8787 8788

CMD ["./backend_app"]
```

Deployment Guide

Single-Node Deployment

System Requirements

- **OS:** Linux, macOS, or Windows
- **RAM:** 512MB minimum, 2GB recommended
- **Disk:** Depends on file size, minimum 100MB
- **Network:** TCP ports 8787, 8788 available

Installation Steps

1. **Compile Backend**

```
cd backend
mkdir build && cd build
cmake ..
make
```

2. **Setup Directories**

```
mkdir -p /var/loopline/{received,sent,shared}
chmod 755 /var/loopline/*
```

3. **Deploy Frontend**

```
cd frontend
npm install
npm run build
# Serve static files from dist/
```

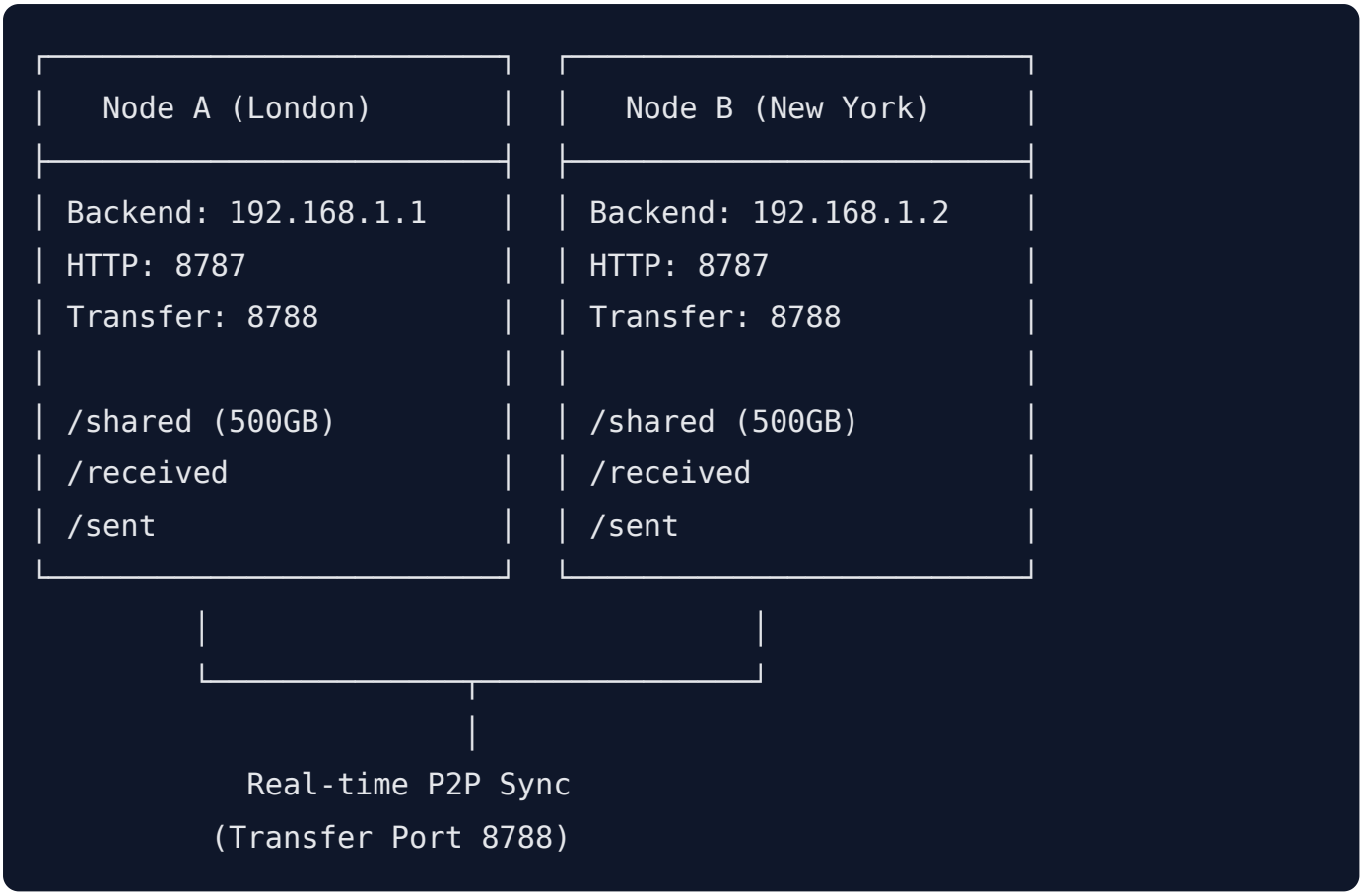
4. **Start Backend**

```
export P2P_HTTP_PORT=8787
export P2P_TRANSFER_PORT=8788
./backend_app
```

5. **Access Frontend**
   - Open browser to `http://localhost:3000`
   - Backend API available at `http://localhost:8787`

Multi-Node Cluster Deployment

Architecture



Configuration for Node A

```
export P2P_HTTP_PORT=8787
export P2P_TRANSFER_PORT=8788
export P2P_BIND_HOST=0.0.0.0
export P2P_ADVERTISED_HOST=node-a.loopline.internal
export P2P_SYNC_PEERS=node-b.loopline.internal:8788
export P2P_SHARED_DIR=/mnt/nfs/shared
./backend_app
```

Configuration for Node B

```
export P2P_HTTP_PORT=8787
export P2P_TRANSFER_PORT=8788
export P2P_BIND_HOST=0.0.0.0
export P2P_ADVERTISED_HOST=node-b.loopline.internal
export P2P_SYNC_PEERS=node-a.loopline.internal:8788
export P2P_SHARED_DIR=/mnt/nfs/shared
./backend_app
```

Kubernetes Deployment

StatefulSet Configuration

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: loopline-backend
spec:
  serviceName: loopline-backend
  replicas: 3
  selector:
    matchLabels:
      app: loopline-backend
  template:
    metadata:
      labels:
        app: loopline-backend
    spec:
      containers:
      - name: backend
        image: loopline-backend:latest
        ports:
        - containerPort: 8787
          name: http
        - containerPort: 8788
          name: transfer
        env:
        - name: P2P_HTTP_PORT
          value: "8787"
        - name: P2P_TRANSFER_PORT
          value: "8788"
        - name: P2P_ADVERTISED_HOST
          valueFrom:
            fieldRef:
              fieldPath: status.podIP
        - name: P2P_SYNC_PEERS
          value: "loopline-backend-0.loopline-backend:8788,loopline-
backend-1.loopline-backend:8788,loopline-backend-2.loopline-
backend:8788"
        volumeMounts:
```

```
        - name: shared
          mountPath: /shared
        - name: received
          mountPath: /backend/received
  volumeClaimTemplates:
  - metadata:
      name: shared
    spec:
      accessModes: [ "ReadWriteOnce" ]
      resources:
        requests:
          storage: 500Gi
  - metadata:
      name: received
    spec:
      accessModes: [ "ReadWriteOnce" ]
      resources:
        requests:
          storage: 100Gi
```

Service Configuration

```yaml
apiVersion: v1
kind: Service
metadata:
  name: loopline-backend
spec:
  clusterIP: None
  ports:
  - port: 8787
    name: http
  - port: 8788
    name: transfer
  selector:
    app: loopline-backend
---
apiVersion: v1
kind: Service
metadata:
  name: loopline-backend-lb
spec:
  type: LoadBalancer
  ports:
  - port: 80
    targetPort: 8787
    name: http
  - port: 8788
    targetPort: 8788
    name: transfer
  selector:
    app: loopline-backend
```

Monitoring & Logging

Metrics to Monitor

```
HTTP Server:
- Port 8787 availability
- Active connections count
- Request latency (p50, p95, p99)
- Error rate (4xx, 5xx)

Transfer Server:
- Port 8788 availability
- Active P2P connections
- Throughput (bytes/sec)
- Transfer success rate

Services:
- FileVaultService: Disk I/O, directory scans
- TransferService: Upload/download counts
- SharedSyncService: Sync lag, change detection

System:
- CPU usage
- Memory usage
- Disk usage (per directory)
- Network I/O
```

Health Check Endpoint (Recommended Addition)

```
GET /health
Response:
{
  "status": "healthy",
  "uptime_seconds": 3600,
  "active_connections": 12,
  "transfer_port_open": true,
  "sync_peers_connected": 2,
  "disk_usage": {
    "received": "45%",
    "sent": "12%",
    "shared": "78%"
  }
}
```

Troubleshooting

Common Issues

HTTP Server Won't Start

Error: `Could not start HTTP server on 0.0.0.0:8787`

Causes:

- Port already in use
- No permission to bind to port
- Network interface not available

Solution:

```
# Check port usage
lsof -i :8787  # macOS/Linux
netstat -ano | findstr :8787  # Windows

# Change port
./backend_app --http 9999
```

Peers Not Syncing

Symptoms: Shared folder not updating on remote nodes
Causes:

- Firewall blocking port 8788
- Incorrect peer address
- Network connectivity issue

Debug:

```
# Test connectivity
telnet peer-address 8788  # Check if reachable
ping peer-address        # Check network

# Check peer list
curl http://localhost:8787/api/peers  # Get configured peers
```

File Transfers Failing

Symptoms: Files stuck in upload/download
Causes:

- Disk space full
- File permissions incorrect
- Transfer timeout

Solution:

```
# Check disk space
df -h /var/loopline/received

# Check permissions
chmod 755 /var/loopline/received
chmod 755 /var/loopline/sent
chmod 755 /var/loopline/shared

# Restart backend
kill $(pgrep backend_app)
./backend_app
```

Frontend Can't Connect to Backend

Error: Connection refused or CORS errors
Causes:

- Backend not running
- Wrong backend URL in frontend config
- CORS policy issues

Solution:

```
# Verify backend is running
curl http://localhost:8787/api/status

# Check frontend configuration
cat .env | grep REACT_APP_BACKEND_URL

# Add CORS headers to backend (if needed)
# Modify http_view.hpp to include:
# Access-Control-Allow-Origin: *
```

High Memory Usage

Symptoms: Backend process consuming excessive RAM
Causes:

- Large files in memory
- Memory leak in service
- Too many concurrent connections

Debug:

```
# Monitor process
top -p $(pgrep backend_app)
ps aux | grep backend_app

# Check open files
lsof -p $(pgrep backend_app)

# Reduce concurrent connections
# Add connection pool limit in http_server()
```

Disk Space Issues

Symptoms: Files not transferring, directories not created
Causes:

- Disk full
- Wrong mount point
- Permission denied

Solution:

```
# Check disk usage
du -sh /var/loopline/*

# Clean old files
find /var/loopline/received -mtime +30 -delete

# Change storage location
export P2P_RECEIVED_DIR=/mnt/larger_disk/received
```

Debug Mode

Enable verbose logging:

```
# Windows
set DEBUG=loopline:*
.\backend_app.exe

# Linux/macOS
export DEBUG=loopline:*
./backend_app
```

Performance Tuning

Recommended Settings by Use Case

Small Office (< 10 users, < 1TB data)

```
P2P_HTTP_PORT=8787
P2P_TRANSFER_PORT=8788
P2P_BIND_HOST=0.0.0.0
P2P_ALLOW_REMOTE=false  # Restrict to local network
```

Medium Enterprise (10-100 users, 1-10TB data)

```
P2P_HTTP_PORT=8787
P2P_TRANSFER_PORT=8788
P2P_BIND_HOST=0.0.0.0
P2P_ALLOW_REMOTE=true
# Deploy on VM with 4GB RAM, 500GB SSD
```

Large Deployment (> 100 users, > 10TB data)

```
# Use Kubernetes cluster
# Replicas: 5+
# Storage: NFS mount for shared folder
# Load balancer: Nginx or cloud provider
```

Conclusion

This documentation provides a complete technical reference for the Loopline P2P file transfer and synchronization system. It covers:

- Backend architecture and components
- Frontend design and implementation
- API specifications
- Data flow and synchronization
- Configuration options
- Deployment strategies
- Troubleshooting guides
- Performance tuning

For updates, refer to the source code and commit history. Report issues and feature requests through the project's issue tracker.

Document Version: 1.0
Last Updated: May 25, 2026
Maintained By: Development Team